

Use index notation, powers and roots

Step 44

Represent repeated multiplication efficiently and connect it to inverse ideas.

★ Today I am learning to use index notation, powers and roots.

1. Write as an Index (Power)

Write each repeated multiplication using index notation.

a) $2 \times 2 \times 2 =$ _____

e) $10 \times 10 \times 10 =$ _____

i) $4 \times 4 \times 4 \times 4 \times 4 =$ _____

b) $5 \times 5 \times 5 \times 5 =$ _____

f) $a \times a \times a \times a =$ _____

j) $b \times b \times b =$ _____

c) $3 \times 3 \times 3 \times 3 \times 3 =$ _____

g) $x \times x \times x \times x \times x \times x =$ _____

k) $6 \times 6 \times 6 \times 6 \times 6 \times 6 \times 6 =$ _____

d) $7 \times 7 =$ _____

h) $9 \times 9 \times 9 \times 9 =$ _____

l) $y \times y =$ _____

Remember:

An index (power) shows how many times the base is multiplied by itself.

Example:

$2 \times 2 \times 2 = 2^3$

(2 is multiplied by itself 3 times.)

2. Expand the Index

Write each index (power) as repeated multiplication.

a) $2^3 =$ _____

e) $10^2 =$ _____

i) $9^2 =$ _____

b) $4^2 =$ _____

f) $a^3 =$ _____

j) $b^2 =$ _____

c) $5^3 =$ _____

g) $x^4 =$ _____

k) $y^5 =$ _____

d) $3^4 =$ _____

h) $7^3 =$ _____

l) $6^4 =$ _____

Remember:

Expanding the index means writing the multiplication in full.

Example:

$3^4 = 3 \times 3 \times 3 \times 3$

3. Evaluate the Powers

Evaluate each power.

a) $2^3 =$ _____

e) $10^2 =$ _____

i) $7^2 =$ _____

m) $8^2 =$ _____

b) $3^2 =$ _____

f) $2^4 =$ _____

j) $5^3 =$ _____

n) $4^4 =$ _____

c) $5^2 =$ _____

g) $6^2 =$ _____

k) $9^2 =$ _____

o) $10^3 =$ _____

d) $4^3 =$ _____

h) $3^5 =$ _____

l) $2^6 =$ _____

p) $3^3 =$ _____

Remember:

Use multiplication to find the value.

Example:

$2^3 = 2 \times 2 \times 2 = 8$

4. Roots (Square Roots)

Find the square root of each number.

a) $\sqrt{4} =$ _____

e) $\sqrt{36} =$ _____

i) $\sqrt{100} =$ _____

b) $\sqrt{9} =$ _____

f) $\sqrt{49} =$ _____

j) $\sqrt{121} =$ _____

c) $\sqrt{16} =$ _____

g) $\sqrt{64} =$ _____

k) $\sqrt{144} =$ _____

d) $\sqrt{25} =$ _____

h) $\sqrt{81} =$ _____

l) $\sqrt{169} =$ _____

Remember:

The square root asks: 'What number multiplied by itself gives this number?'

Example: $\sqrt{25} = 5$ because $5 \times 5 = 25$

5. Connect Powers and Roots

Use powers and roots to complete the number sentences.

a) $(\sqrt{49})^2 =$ _____

e) $(\sqrt{121})^2 =$ _____

i) $\sqrt{5^2} =$ _____

b) $\sqrt{6^2} =$ _____

f) $(\sqrt{15})^2 =$ _____

j) $(\sqrt{9})^2 =$ _____

c) $(\sqrt{8})^2 =$ _____

g) $\sqrt{13^2} =$ _____

k) $3^2 =$ _____ $\rightarrow \sqrt{\quad} =$ _____

d) $\sqrt{7^2} =$ _____

h) $(\sqrt{20})^2 =$ _____

l) $6^2 =$ _____ $\rightarrow \sqrt{\quad} =$ _____

Remember:

Powers and roots are inverses (opposites). They undo each other.

Example: $(\sqrt{9})^2 = 9$ and $\sqrt{3^2} = 3$

6. Problem Solving

Solve the problems using powers and roots.

- a) A bacteria cell splits in two every hour. If there is 1 cell at the start, how many cells are there after 5 hours?

Write your answer as a power of 2 and then find the value.

Answer: _____



- b) The area of a square garden is 144 m². What is the length of one side of the garden?

Answer: _____



- c) A cube has a volume of 216 cm³. What is the length of one edge of the cube?

Answer: _____



- d) A number squared is 169. What is the number?

Answer: _____



My Learning Check

Circle how confident you feel.



I can do it!



Almost there!



I need more help.



This was hard.

Parent Observation (optional)

Please tick the boxes that apply.

- | | |
|--|---|
| ☐ Understands index notation | ☐ Connects powers and roots |
| ☐ Expands powers correctly | ☐ Solves problems using powers/roots |
| ☐ Evaluates powers accurately | ☐ Shows effort and persistence |
| ☐ Understands square roots | ☐ Works neatly and carefully |

Notes: _____

Today I learned...

1 thing I did well: _____

1 thing I will improve: _____

